

Biology Midterm — Student A

Q1: Describe the structure and function of the cell membrane.

The plasma membrane acts as a selective barrier composed of a lipid bilayer, allowing certain molecules through while blocking others, and contains proteins involved in transport and communication.

Q2: Explain the role of DNA in a cell.

DNA holds the hereditary information necessary for an organism's development and function, stored as coded sequences of nucleotides grouped into genes.

Q3: Describe the process of photosynthesis.

Plants use photosynthesis to transform sunlight, carbon dioxide, and water into glucose and oxygen, a process that takes place in chloroplasts with the help of chlorophyll.